

## **Tutorial 39**

M1 verdicts — and the manager wedge, by hand

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Advanced Corporate Finance (EBM232A05)

University of Groningen

- **First: assessment moment M1.** Your scoping memos come back graded (10 points) with the proceed / no-proceed verdict. Twenty minutes on what the verdicts mean and the pattern in them.
- **Then: Problem set 39** — the manager wedge, all four problems at the board.

Also this week: the **mock exam** is posted on Brightspace. Attempt it in Week 40 — it is the calibration for the real one.

## **M1 — the verdicts**

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## What the verdict means

- **Proceed**: your target cells, variable definitions and data plan are precise enough to build on. Start the pull now if you have not.
- **Revise and proceed**: something is not yet implementable — usually a **variable definition a stranger could not code from**. Resubmit within one week; the M1 score stands, but an ignored revision costs the same points again at M2.
- The score (10): parameter-sheet match (1) · claims list (2) · **variable-definition table (4)** · data plan (2) · repo skeleton (1).

The pattern across teams, every year: the four points on the variable-definition table are where memos differ. Vagueness there is the single best predictor of a failed replication — an ambiguity found now is a five-minute fix; found in Week 42 it is fatal.

## Between now and M2 (week 41)

- **Week 40 is Career Week** — no lecture, no tutorial, and you will be busy with events. Plan accordingly: the reproduction should be **running before** Career Week, so week 40 is buffer, not the main push.
- M2 (due 22:00 Tuesday of week 41): target cells against the paper's printed values, plus the **sample funnel** and reconciliation memo. 25 points.
- Keep the dossier live **as you work**: catches, transcript references, commits. Reconstructing it later is visible — and graded accordingly.

## **Problem set 39, discussed**

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The contract is  $w = \alpha + \beta y$ . A project adds  $G > 0$  to the expected outcome and  $\Delta\sigma^2$  to its variance; the manager's certainty equivalent is

$$CE(\beta) = \beta G - k\beta^2, \quad k \equiv \frac{r}{2}\Delta\sigma^2.$$

- (a) Why is the gain linear in  $\beta$  but the risk cost quadratic?
- (b) Derive his investment rule and the refusal cutoff  $\beta^*$ .
- (c) Where does his *enthusiasm* peak? Show it is half the cutoff.
- (d) State the wedge: whose hurdle, against whose?

$$\text{take} \iff G \geq k\beta \iff \beta \leq \beta^* = \frac{G}{k}, \quad \text{enthusiasm peaks at } \hat{\beta} = \frac{G}{2k} = \frac{\beta^*}{2}.$$

- Share of the payoff  $\Rightarrow$  gain  $\propto \beta$ ; variance of his pay  $\propto \beta^2$  — linear against quadratic is the entire model.
- The hurdle  $k\beta$  **rises** in his stake: stronger incentives make him *more* conservative. “More equity, more alignment” holds only below  $\hat{\beta}$ .
- The wedge: shareholders test  $G \geq 0$  (diversified); he tests  $G \geq k\beta$ . The gap is **manufactured by the incentive scheme itself**.

$G = 12, k = 80$ .

| Manager's stake | His share $\beta G$ | His risk cost $k\beta^2$ | $CE(\beta)$ |
|-----------------|---------------------|--------------------------|-------------|
| $\beta = 0.05$  | ?                   | ?                        | ?           |
| $\beta = 0.075$ | ?                   | ?                        | ?           |
| $\beta = 0.15$  | ?                   | ?                        | ?           |
| $\beta = 0.25$  | ?                   | ?                        | ?           |

Then: the refusal cutoff; the two dials ( $k \rightarrow 40$  vs.  $G \rightarrow 24$ ); and the CFO's memo — “raise  $\beta$  from 0.075 to 0.25 to strengthen incentives”.

| Manager's stake | $\beta G$ | $k\beta^2$ | $CE(\beta)$     |
|-----------------|-----------|------------|-----------------|
| $\beta = 0.05$  | 0.6       | 0.2        | +0.4            |
| $\beta = 0.075$ | 0.9       | 0.45       | + <b>0.45</b> ↑ |
| $\beta = 0.15$  | 1.8       | 1.8        | 0               |
| $\beta = 0.25$  | 3.0       | 5.0        | -2.0 ↓          |

- $\beta^* = 12/80 = \mathbf{0.15}$ ; peak at 0.075. Both dials give  $\beta^* = \mathbf{0.30}$  — the cutoff is the **ratio**  $G/k$ .
- The CFO's proposal crosses the cutoff: the CEO refuses the class, forgoing  $G = 12$ . **Incentive strength is not monotone in  $\beta$ .**

Outcome  $y \in \{80, 120\}$ , equally likely. Stock pays  $\beta y$ ; an option pays  $\max\{y - 100, 0\}$ .

- (a) Value one option. Spread the outcome to  $\{60, 140\}$  (same mean): what happens to stock, and to the option? Name the property.
- (b) A project changes  $y$  to  $\{55, 135\}$  — mean down 5, spread up. Shareholders? Stock-paid manager? Option-paid manager?
- (c) Who pays for the option-holder's verdict, and through which channel?
- (d) The board's trade-off in one sentence.

- (a) Option:  $\frac{1}{2} \times 20 = 10$ ; after the spread **20**, stock **untouched**. The property: **convexity** — vega; a linear claim has none.
- (b) Mean falls 5: shareholders **no**; stock-manager ( $-5\beta$ ) **no**; option-manager ( $17.5 > 10$ ) **yes**. **One project, three verdicts — the sign depends only on the curvature of the claim.**
- (c) The loss lands on the **fixed claimants**: the bad state worsened ( $80 \rightarrow 55$ ). Asset substitution, manufactured by the **pay package** — and in Week 42, the same truncation is held by the deposit insurer.
- (d) Convexity buys back risk appetite — and manufactures appetite for value-destroying variance. **No setting removes the wedge; settings trade it.**

- (a) Jensen and Murphy (1990): \$3.25 per \$1,000 —  $\beta \approx 0.003$ . When is that small number a *choice*, not a puzzle?
- (b) What would you measure about the manager to decide between “captured boards” and “optimal risk-sharing”?
- (c) Coles et al. (2006) vs. Bakke et al. (2016): which parts of P3 do they test, and why is the second design cleaner?
- (d) The Week-39 ledger row.

- (a) With  $k$  large — an undiversified manager — even small  $\beta$  loads real risk;  $\beta \leq \beta^* = G/k$  makes \$3.25 what an *efficient* contract looks like. The puzzle exists only against the  $\beta = 1$  benchmark.
- (b)  $k$  itself: wealth and human capital in the firm, hedgeability, horizon. **The same number is evidence for either story until  $k$  is measured.**
- (c) Both test the convexity dial. Cross-section (Coles et al., 2006): vega and risky policy co-move — but **pay is endogenous**. Bakke et al. (2016) use variation in option pay the firm did not choose for risk reasons: hedging falls — the P3 comparative static with a causal sign.

## The Week-39 ledger row

| Wk | Friction                                        | Effect                                                               | Evidence                                                                     |
|----|-------------------------------------------------|----------------------------------------------------------------------|------------------------------------------------------------------------------|
| 39 | Manager $\neq$ shareholder (risk-bearing wedge) | risk-taking $\downarrow$ in $\beta$ ; $\uparrow$ in vega (overshoot) | (Jensen and Murphy, 1990); (Coles et al., 2006); causal (Bakke et al., 2016) |

Same row as the lecture's close — the set's contribution: you can now **locate the cutoff** instead of quoting the sign.

## Before you leave

- **Career Week next week** — no sessions. The reproduction should be running *before* it; week 40 is buffer.
- **M2 due 22:00 Tuesday, week 41**: target cells + sample funnel + reconciliation. Keep logging catches as they happen.
- **Mock exam** is up — attempt it under exam conditions during week 40.
- **Problem set 39 key** on Brightspace after today; the manager-wedge Explorer is the self-check.

- Bakke, T.-E. et al. (2016). **“The Causal Effect of Option Pay on Corporate Risk Management”**. In: *Journal of Financial Economics* 120.3, pp. 623–643.
- Coles, J. L., N. D. Daniel, and L. Naveen (2006). **“Managerial Incentives and Risk-Taking”**. In: *Journal of Financial Economics* 79.2, pp. 431–468.
- Jensen, M. C. and K. J. Murphy (1990). **“Performance Pay and Top-Management Incentives”**. In: *Journal of Political Economy* 98.2, pp. 225–264.